CS 402 – Project 3 Report

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Overview

In this project, I implemented solutions to several problems involving trees and dynamic programming in C++. The goal was to follow the given specifications exactly and ensure the code worked correctly with the autograder.

Problem 1: Weird Tree Traversal

For this problem, the tree was given using a first-child, next-sibling representation. I performed a single breadth-first traversal while keeping track of each node's depth. Nodes at even depths were collected in order, while nodes at odd depths were stored separately. After the traversal finished, the odd-depth nodes were added in reverse order to produce the required traversal.

Problem 2: Bits-to-Tree

This problem required rebuilding a tree from a sequence of bits representing a DFS traversal. I first checked whether the input was valid. I then used a stack to construct the tree. A `1` created a new child node, and a `0` moved back to the parent. Nodes were labeled in preorder as required.

Problem 3: Light Post Problem

To solve this problem, I treated the graph as a tree and used dynamic programming. For each node, I computed the minimum cost for different lighting cases. A depth-first search was used to compute the costs, and another traversal was used to determine which nodes should contain light posts.

Problem 4: Smallest Subset Sum

I used dynamic programming to find a subset that sums to the target value using the fewest elements. To ensure the subset was lexicographically smallest, I only updated the solution when a strictly smaller subset was found. After filling the table, I reconstructed the subset by backtracking.

Testing

I tested the code using small test cases and compiled it with strict compiler flags to ensure there were no warnings or errors. All test code was removed before submission.

Conclusion

This project helped me practice tree traversal techniques and dynamic programming. The final code follows all requirements and works correctly with the provided tests.